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Sakshi Matale

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EDUCATION

Jai Hind College, Mumbai, India (March 2025)

Secured a Bachelor of Science in Physics with Distinction (CGPA 9.18 / 10.0) (1.4 in German grading system)

Mithibai College, Mumbai, India (March 2022)

Secured Distinction (65.50%) in Higher Secondary Certificate Education

PROFESSIONAL EXPERIENCE

WeMR, Mumbai,India, Research Intern, (November, 2024 - February 2025)

Completed a research internship in The Standard Model Lagrangian- Literature Review for the duration of 4 months under the guidance of Dr.Namrataa Kkommineni organized by WeMR, the skill Foundation.Conducted an in-depth literature review on the Standard Model Lagrangian, analyzing peer-reviewed research papers and advanced theoretical texts.

- Studied and interpreted key concepts including gauge symmetry, spontaneous symmetry breaking, Higgs mechanism, QCD, and electroweak unification.
- Analyzed the mathematical structure and interaction terms of the Standard Model Lagrangian using group theory and quantum field theory foundations.
- Prepared and delivered technical presentations explaining complex particle physics concepts with structured theoretical derivations.
- Engaged in regular academic discussions to clarify theoretical interpretations and strengthen conceptual understanding.
- Explored applications of machine learning in high-energy physics data analysis, connecting theoretical physics with data-driven approaches.

ACADEMIC PROJECTS AND PAPERS

Movie Recommendation System (Python, Machine Learning)

- Designed and implemented a content-based movie recommendation system using Python.
- Applied data preprocessing, feature extraction, and similarity metrics to recommend movies based on user preferences.
- Worked with real-world datasets and improved recommendation accuracy using vectorization techniques.

House Price Prediction Model (Machine Learning)

- Built a basic regression-based house price prediction model using Python.
- Performed data cleaning, exploratory data analysis (EDA), and feature selection.
- Evaluated model performance using appropriate error metrics and improved predictions through tuning.

Python Mini Projects (Core Programming & Logic Building)

Developed multiple Python-based applications including:

- Calculator
- Higher or Lower Game
- Password Generator

- Pizza Ordering System
- Rock–Paper–Scissors Game
- Number Guessing Game
- Coffee Machine Simulation
- Blind Auction System

Strengthened understanding of control structures, functions, loops, conditional logic, and user input handling. Focused on writing clean, modular, and reusable code.

TECHNICAL SKILLS

Languages and Programming Skills

- Programming: Python (NumPy, Pandas), C++
- Machine Learning: Regression, Classification, Clustering
- Data Analysis using NumPy and Pandas, Data Cleaning
- Other Tools: Jupyter Notebook, Excel, LaTeX (for academic writing), Basics of Web Development

EXTRA-CURRICULAR ACTIVITIES

Courses and Certificates

- Physics Society Committee Member — [Jai Hind College]. Volunteered as part of the organizing committee, managing academic events, student exhibitions, and physics outreach activities.
- Attended 3 days program in Astronomy organized by Khagol Mandal with IAU from 1st to 3rd December 2023 at IITB, Mumbai.
- 100 Days of Code: Python Bootcamp — Udemy (Completed, 2023). Learned Python programming through 100+ projects, covering data structures, OOP, web development, and automation.
- The Ultimate Job-Ready Data Science Course — CodeWithHarry (Ongoing, 2025). Topics: Python for Data Science, data analysis (NumPy, Pandas), visualization (Matplotlib, Seaborn), SQL, probability & statistics, machine learning (scikit-learn), deep learning, and applied projects.
- Selected for 7 days Summer School Program in Theoretical Astrophysics held from 3rd June 2024 to 8th June 2024 at St. Xavier's College, Ahmedabad